

ME 2801

Introduction to Control Systems

HW3 Review

Takeaways

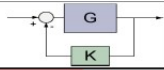
First and second order systems

- Forms of TF's and step response in the t-domain
- Step and Impulse response
- Viewing pole locations in the (complex) s-plane
- Performance metrics

$$G_1(s) = K_{dc} \frac{a}{s+a} = K_{dc} \frac{1}{\tau s + 1}$$

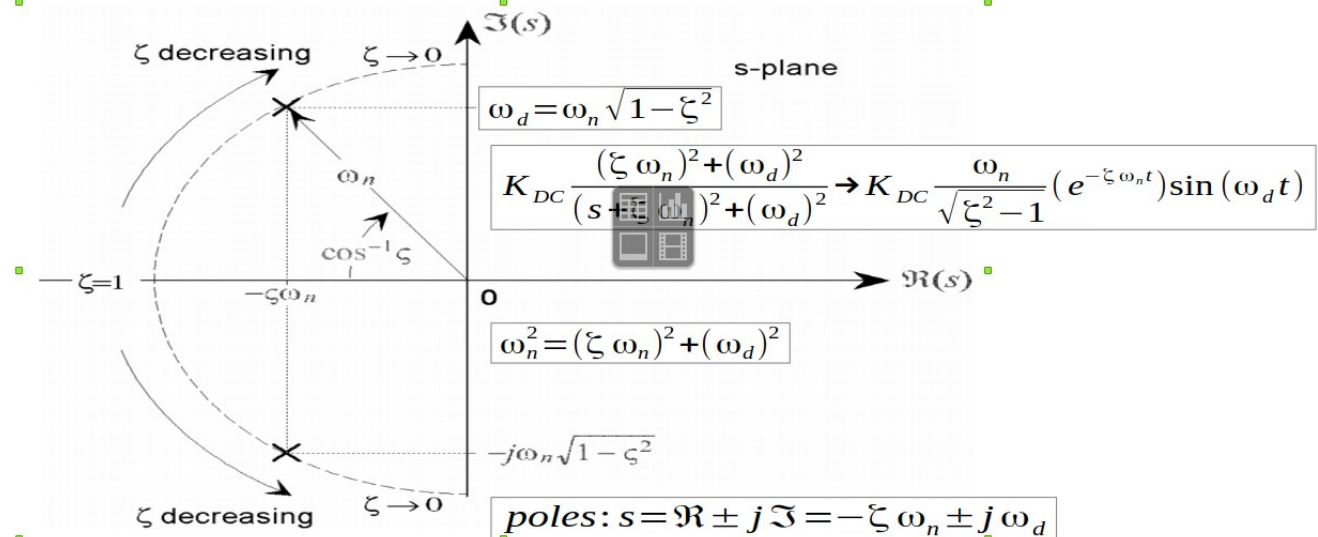


Underdamped: Complex Pole Geometry



Click to add text

$$K_{DC} \frac{\omega_n^2}{(s+\Re)^2 + (\Im)^2} = K_{DC} \frac{(\zeta \omega_n)^2 + (\omega_d)^2}{(s+\zeta \omega_n)^2 + (\omega_d)^2} = K_{DC} \frac{\omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$



21. Repeat Problem 20 using MATLAB. Have the computer program estimate the given specifications and plot the step responses. Estimate the rise time from the plots. [Section: 4.6]

MATLAB

ML

It i view?

```
G1=tf(16,[1 3 16])
```

4.12

12. Write the general form of the capacitor voltage for the electrical network shown in Figure P4.4. [Section: 4.4]

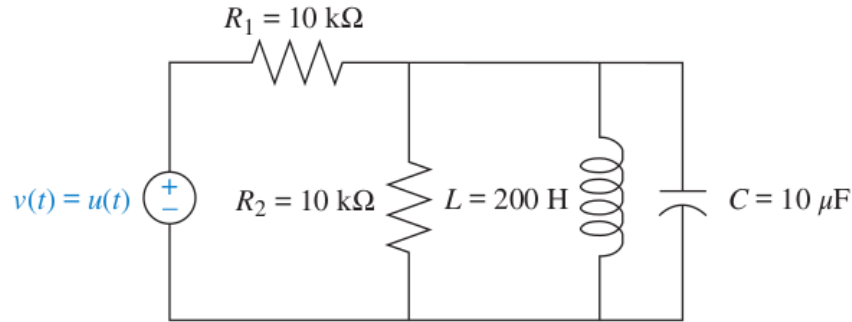


FIGURE P4.4

13. Use MATLAB to plot the capacitor voltage in Problem 12. [Section: 4.4]

MATLAB

ML

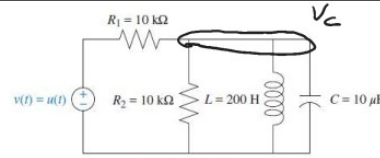


FIGURE P4.4

$$\frac{V_c - V}{R_1} + \frac{V_c}{R_2} + \frac{V_c}{Ls} + V_c(Cs) = 0$$

$$V_c(s) \left(\frac{1}{R_2} + \frac{1}{Ls} + Cs \right) + \frac{V_c(s) - V(s)}{R_1} = 0$$

$$V_c(s) \left(\frac{1}{R_2} + \frac{1}{Ls} + Cs \right) + \frac{V_c(s)}{R_1} - \frac{V(s)}{R_1} = 0$$

$$V_c(s) \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{Ls} + Cs \right) = \frac{V(s)}{R_1}$$

$$\frac{V_c(s)}{V(s)} = \frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{Ls} + Cs} = \frac{10s}{s^2 + 20s + 500}$$

How?

Time Response Performance Metrics

Time Response

$$T_r = \frac{2.2}{a} \quad (4.9); \quad T_s = \frac{4}{a} \quad (4.10)$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (4.22)$$

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 \quad (4.38)$$

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}} \quad (4.39)$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} \quad (4.34); \quad T_s = \frac{4}{\zeta\omega_n} \quad (4.42)$$

Rise Time for 2nd Order Model

* Polynomial approx

$$\omega_n T_r = 1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1$$

max. error less than 1/2% for $0 < \zeta < 0.9$

* Or table lookup

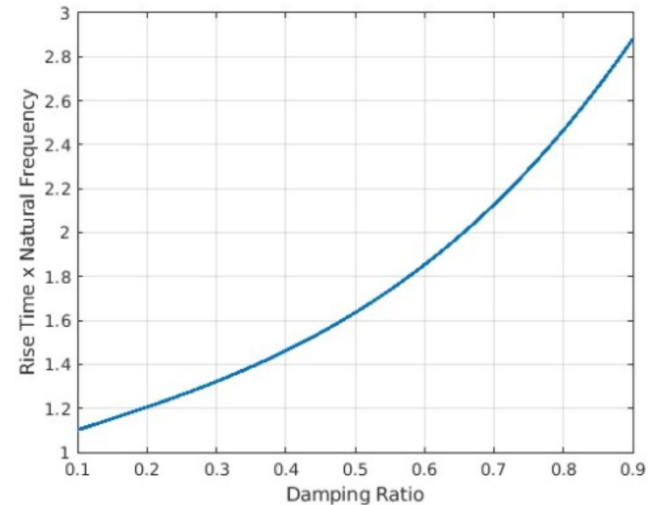


Figure 1: Reproduction of Nise Figure 4.16

Previous Quarters

First and Second Order Models: Step Response

$$c(t) = K[1 - e^{-at}]$$

$$c(t) = K[1 - Ae^{-\sigma_1 t} - Be^{-\sigma_2 t}]$$

$$c(t) = K[1 - Ae^{-\sigma_d t} \cos(\omega_d t - \phi)]$$

$$c(t) = K[1 - A \cos(\omega_1 t - \phi)]$$

$$c(t) = K[1 - Ae^{-\sigma_1 t} - Bte^{-\sigma_1 t}]$$

Nise 4.0 corr.

a) $c(t) = A + Be^{-2t}$, 1st order

b) $c(t) = A + Be^{-3t} + Ce^{-6t}$, over damped

c) $c(t) = A + Be^{-10t} + Ce^{-20t}$, over damped

d) $c(t) = A + Be^{-3t} \cos(3\sqrt{5}t + \phi)$, underdamped
 I thought $\cos(2\pi t)$?
 notes had $\sin(2\pi t)$?

e) $c(t) = A + B \cos(3t + \phi)$, undamped b/c poles have no real component.

f) $c(t) = A + Be^{-10t} + Ce^{-10t}$, critically damped

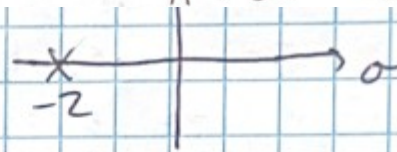
a) $T(s) = \frac{2}{s+2}$

General: $c(t) = K[1 - e^{-at}]$

poles: -2
zeros: none

$\Rightarrow c(t) = A + Be^{-2t}$

First-order response

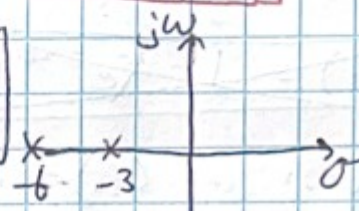


got from solutions.
How do we know it
is first order response?
Can it not be stable
or unstable?

b) $T(s) = \frac{5}{(s+3)(s+6)}$

General: $c(t) = K[1 - Ae^{-\sigma_1 t} - Be^{-\sigma_2 t}]$

poles: -3, -6
zeros: none



First and Second Order Models: Step Response

* the general forms you gave us don't match the format of the solutions (mostly in sign). Not sure where the general solutions came from. Only can find similar example from Skill Assessment Exercise 4.1 on page 161.

* I tried plotting using `step()` and can visualize the stability, but can we discuss how to visualize the poles and zeros?

$$c(t) = K[1 - e^{-at}]$$

$$c(t) = K[1 - Ae^{-\sigma_1 t} - Be^{-\sigma_2 t}]$$

$$c(t) = K[1 - Ae^{-\sigma_d t} \cos(\omega_d t - \phi)]$$

$$c(t) = K[1 - A \cos(\omega_1 t - \phi)]$$

$$c(t) = K[1 - Ae^{-\sigma_1 t} - Bte^{-\sigma_1 t}]$$

8.

a. Pole: -2; $c(t) = A + Be^{-2t}$; first-order response.

b. Poles: -3, -6; $c(t) = A + Be^{-3t} + Ce^{-6t}$; overdamped response.

c. Poles: -10, -20; Zero: -7; $c(t) = A + Be^{-10t} + Ce^{-20t}$; overdamped response.

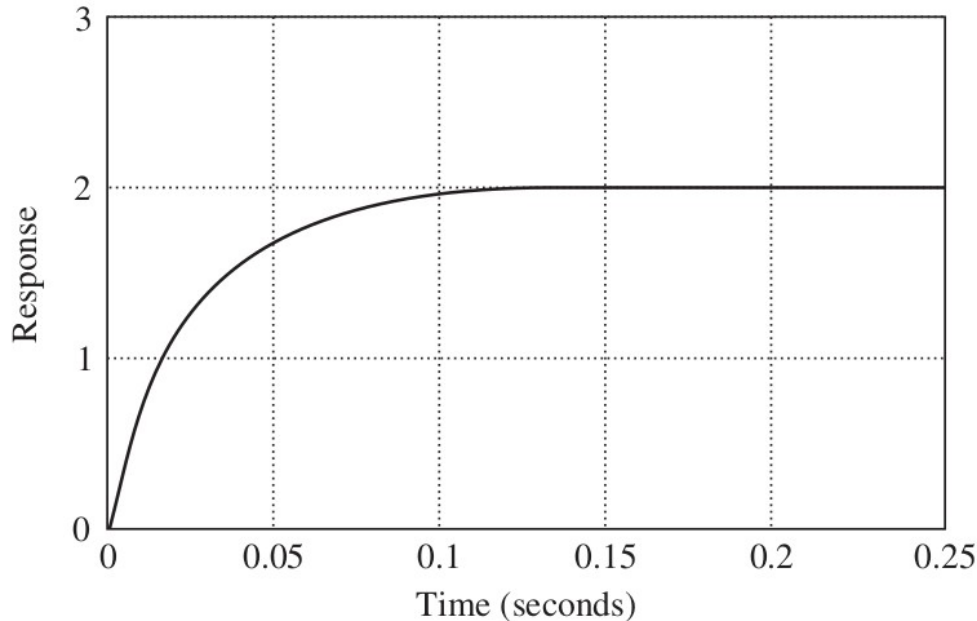
d. Poles: $(-3+j3\sqrt{15})$, $(-3-j3\sqrt{15})$; $c(t) = A + Be^{-3t} \cos(3\sqrt{15}t + \phi)$; underdamped.

e. Poles: $j3$, $-j3$; Zero: -2; $c(t) = A + B \cos(3t + \phi)$; undamped.

f. Poles: -10, -10; Zero: -5; $c(t) = A + Be^{-10t} + Cte^{-10t}$; critically damped.

Nise 4.32

- 32.** For each of the three unit step responses shown in Figure P4.10, find the transfer function of the system.
[Sections: 4.3, 4.6]



Are we *sure* that a first order model generated this response?

This is analogous to Lab 2.

Nise 4.32

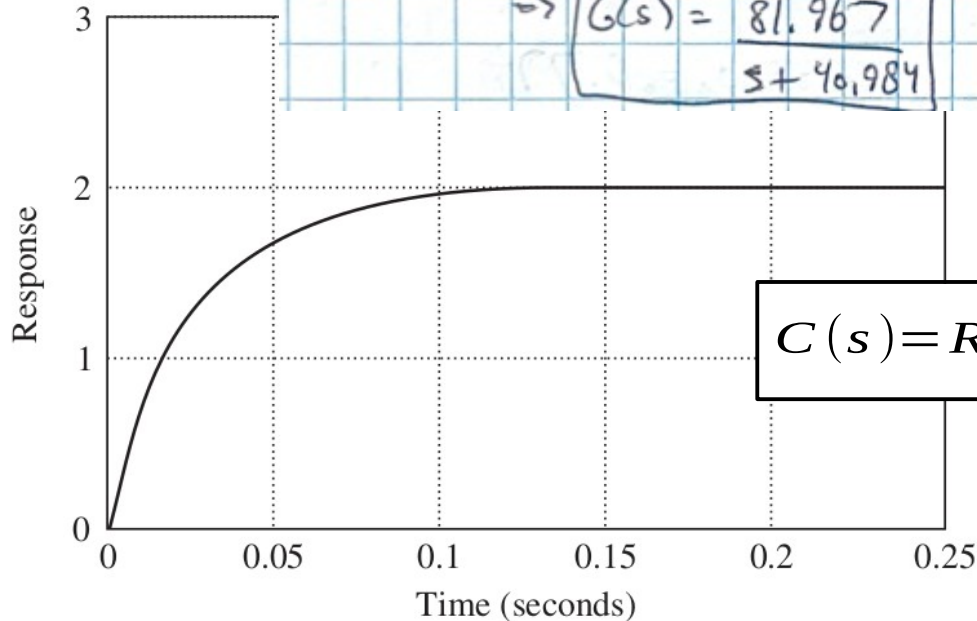
32. For each
Figure F
[Section

* estimating a first order: $G(s) = \frac{K}{s+a}$ but $\frac{K}{a} = 2$

$\Rightarrow \frac{K}{40.984} = 2 \Rightarrow K = 81.967$

$\Rightarrow G(s) = \frac{81.967}{s + 40.984}$

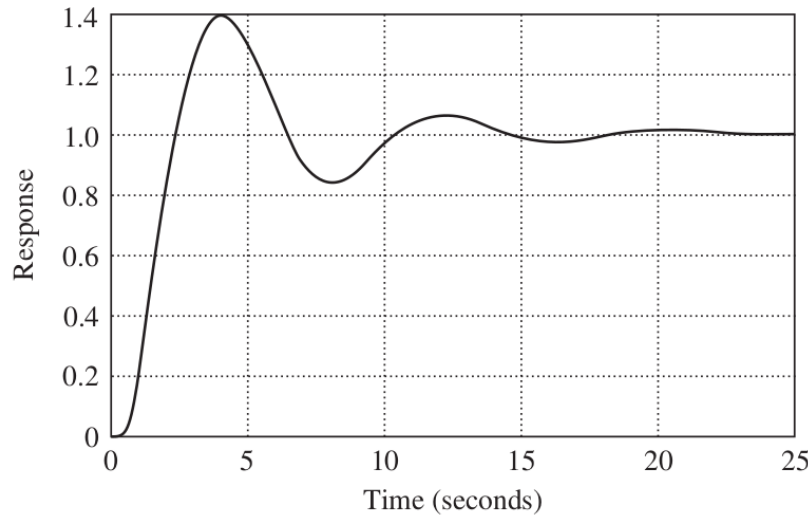
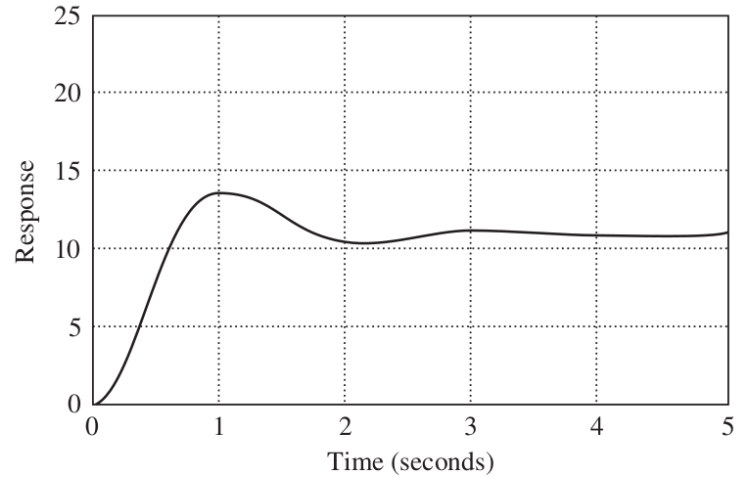
where does this
curve from?
final value?



$$C(s) = R(s)G(s) = K_{DC} \frac{a}{s(s+a)} = K_{DC} \frac{1}{s(\tau s + 1)}$$

$$c(t) = K_{DC} (1 - e^{-t/\tau})$$

Nise 4.32



Identifying 2nd order model:

- * 3 unknowns
- * Choose 3 eqns

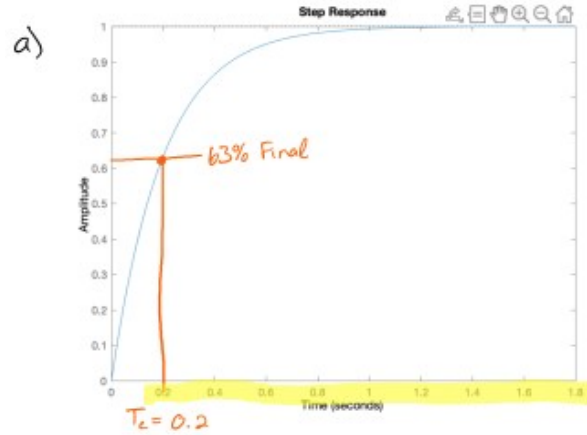
$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 \quad (4.38)$$

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}} \quad (4.39)$$

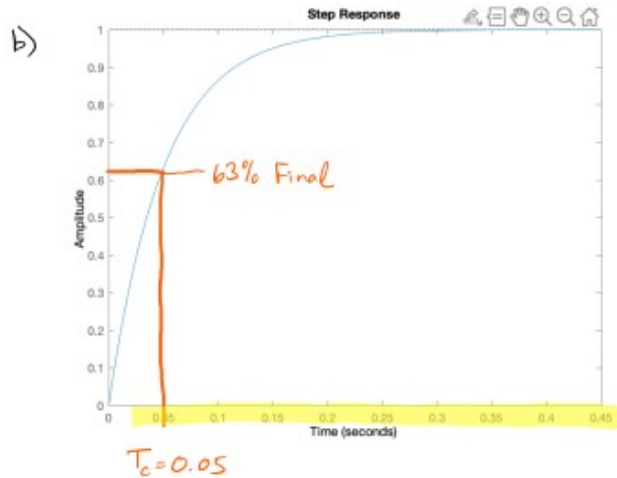
$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} \quad (4.34); \quad T_s = \frac{4}{\zeta\omega_n} \quad (4.42)$$

First-Order Step Response

Nise 4.3.

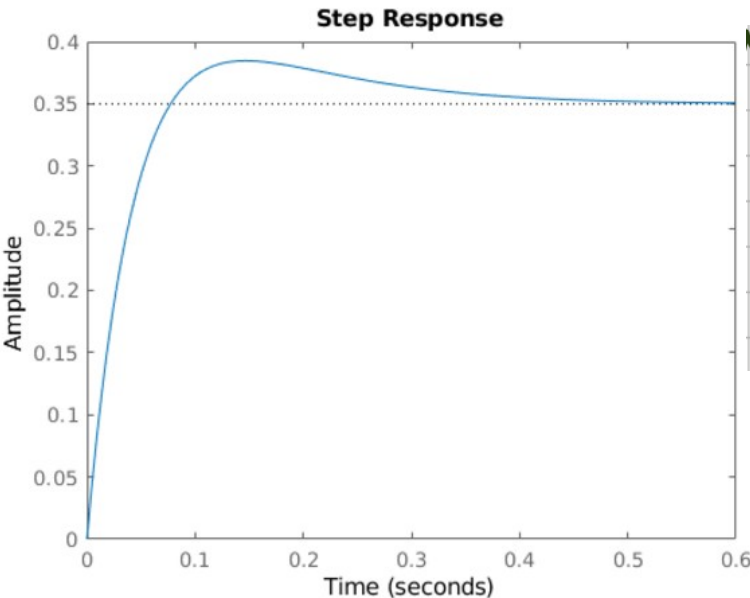
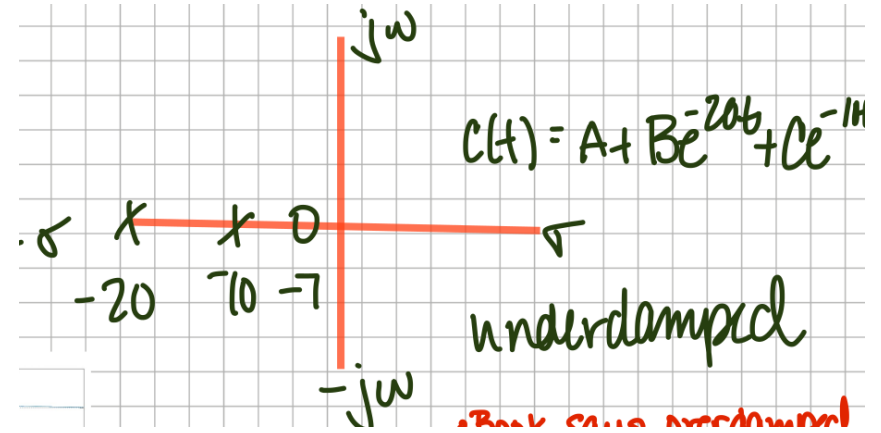


MATLAB plots look the same for these functions except for Time scale.



Effect of Zero 22-3

$$c) \quad T(s) = \frac{10(s+7)}{(s+10)(s+20)}$$



• Book says overdamped
but MATLAB plot looks
underdamped.
over makes sense from
S-plot.



?

$$c(t) = K[1 - e^{-at}]$$

$$c(t) = K[1 - Ae^{-\sigma_1 t} - Be^{-\sigma_2 t}]$$

$$c(t) = K[1 - Ae^{-\sigma_d t} \cos(\omega_d t - \phi)]$$

$$c(t) = K[1 - A \cos(\omega_1 t - \phi)]$$

$$c(t) = K[1 - Ae^{-\sigma_1 t} - Bte^{-\sigma_1 t}]$$

Effect of Zero

Effect of A Zero

Two poles in the denominator

```
den = conv([1 10],[1 20]);
```

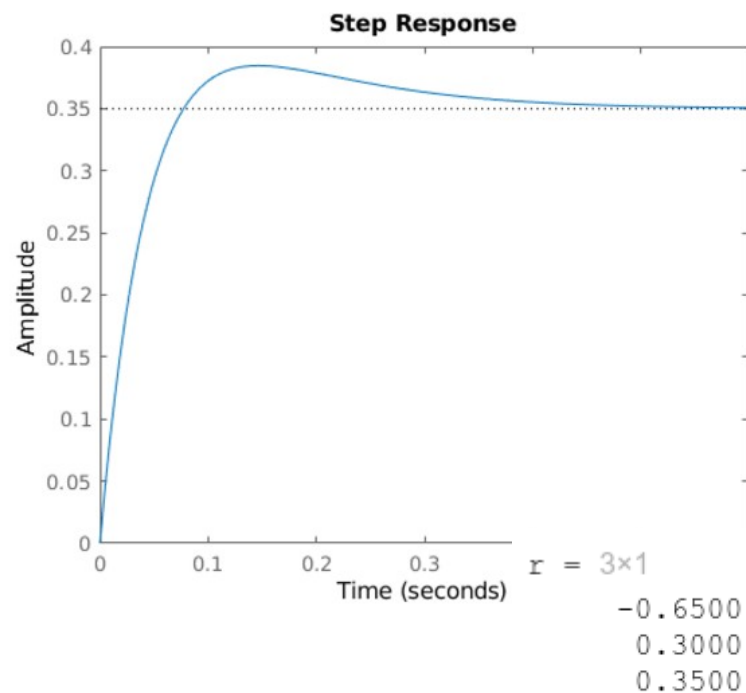
Step response with a zero

```
numwzero = 10*[1 7];  
G = tf(num,den)  
figure(1); clf();  
step(G)
```

Verify the form of the step response with a zero

```
C = tf([1],[1 0])*G  
[cnum, cden] = tfdata(C)  
[r, p, k] = residue(cnum{1}, cden{1})
```

$$c(t) = -0.65e^{-20t} + 0.3e^{-10t} + 0.35$$



$p = 3 \times 1$

-20
-10
0

$k =$

$[]$

Effect of Zero

Effect of A Zero

Two poles in the denominator

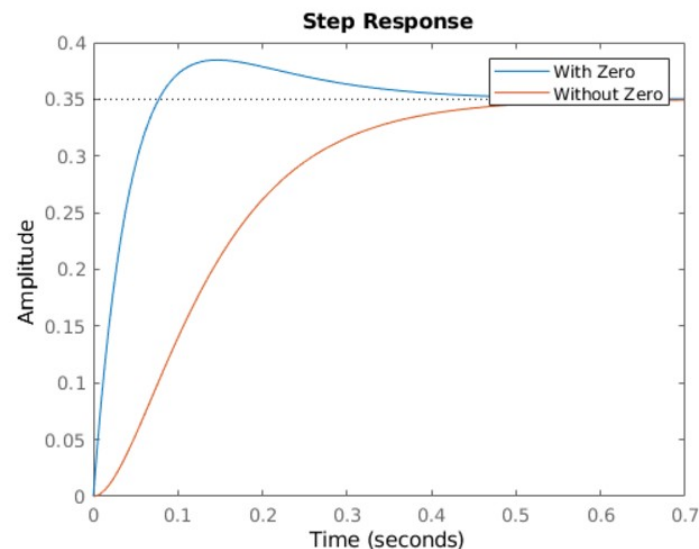
```
den = conv([1 10],[1 20]);
```

Step response with a zero

```
numwzero = 10*[1 7];  
G = tf(num,den)  
figure(1); clf();  
step(G)
```

Overlay step response without a zero

```
G2 = tf(10*7,den)  
hold on  
step(G2)  
legend('With Zero','Without Zero')
```



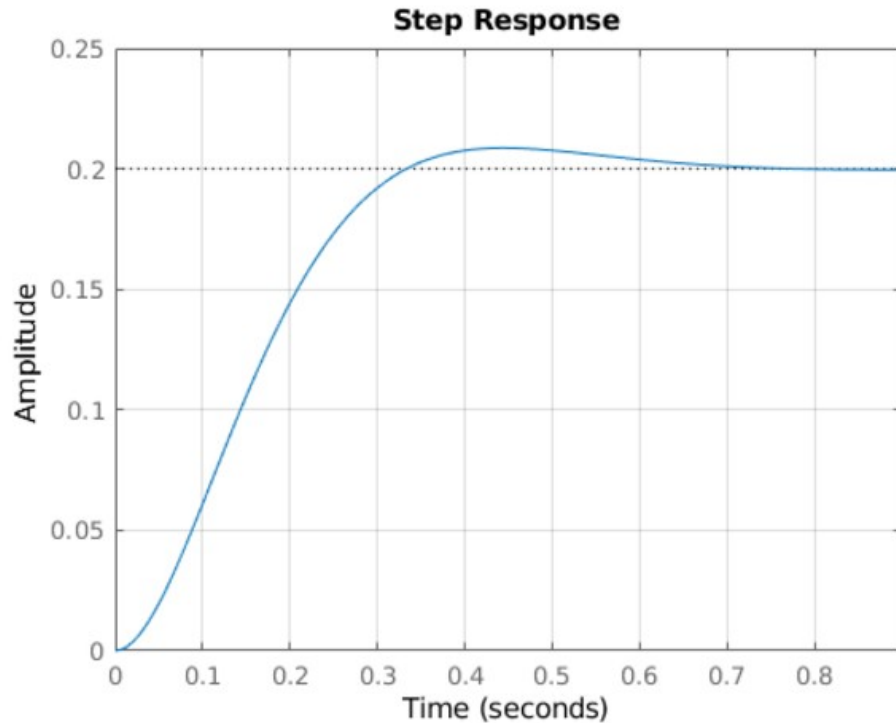
Verify the form of the step response with a zero

```
C = tf([1],[1 0])*G  
[cnum, cden] = tfdata(C)  
[r, p, k] = residue(cnum{1}, cden{1})
```

$$c(t) = -0.65e^{-20t} + 0.3e^{-10t} + 0.35$$

Second Order Model Estimation

Given an unit step response graph



*Hard to check, can we do
b and/or e in class?*

b) }
c) } Not really seeing how to do these backwards

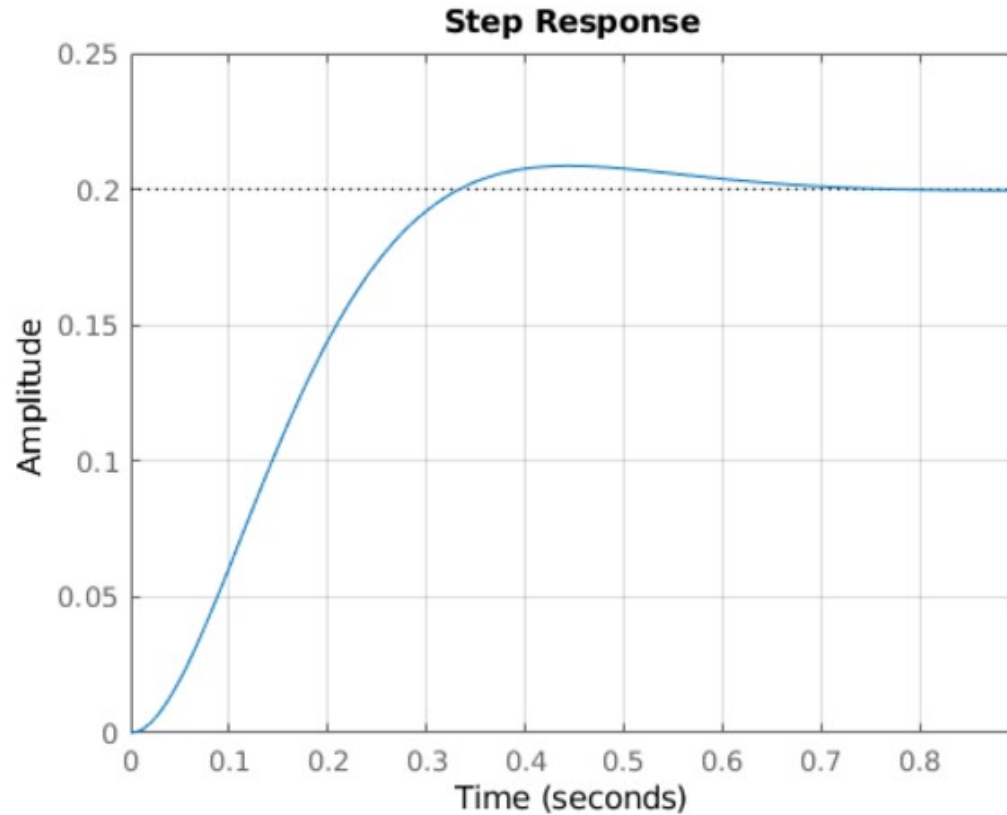
$$G_1(s) = K_{dc} \frac{a}{s+a} = K_{dc} \frac{1}{\tau s + 1}$$

$$G_2(s) = K_{dc} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = K_{dc} \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_d^2}$$

Estimate a transfer function that would generate a similar unit step response graph.

Second Order Model Estimation

Given an unit step response graph



2nd Order TF Model: **Three** Unknowns

$$G_2(s) = K_{dc} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = K_{dc} \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_d^2}$$

Observable Metrics: **Four** Equations

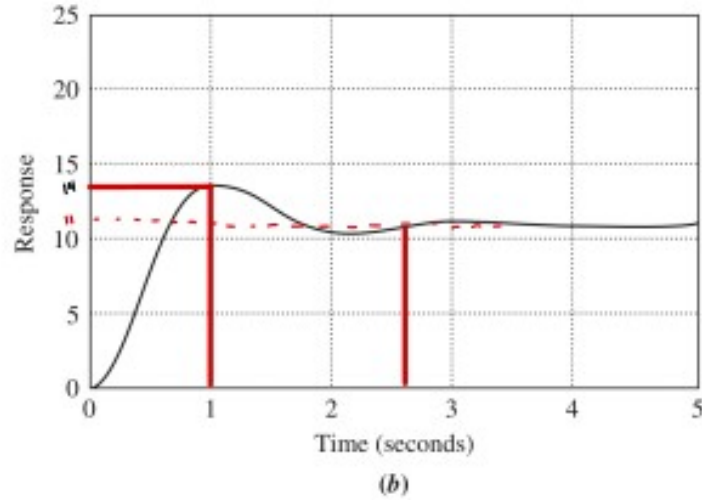
$$K_{dc} = \lim_{\omega \rightarrow 0} [|G(j\omega)|] = G(s = 0)$$

$$\%OS = e^{-(\zeta\pi/\sqrt{1-\zeta^2})} \times 100 \quad (4.38)$$

$$T_s = \frac{4}{\zeta\omega_n} \quad (4.42)$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \quad (4.34)$$

Aside – Estimating Kdc



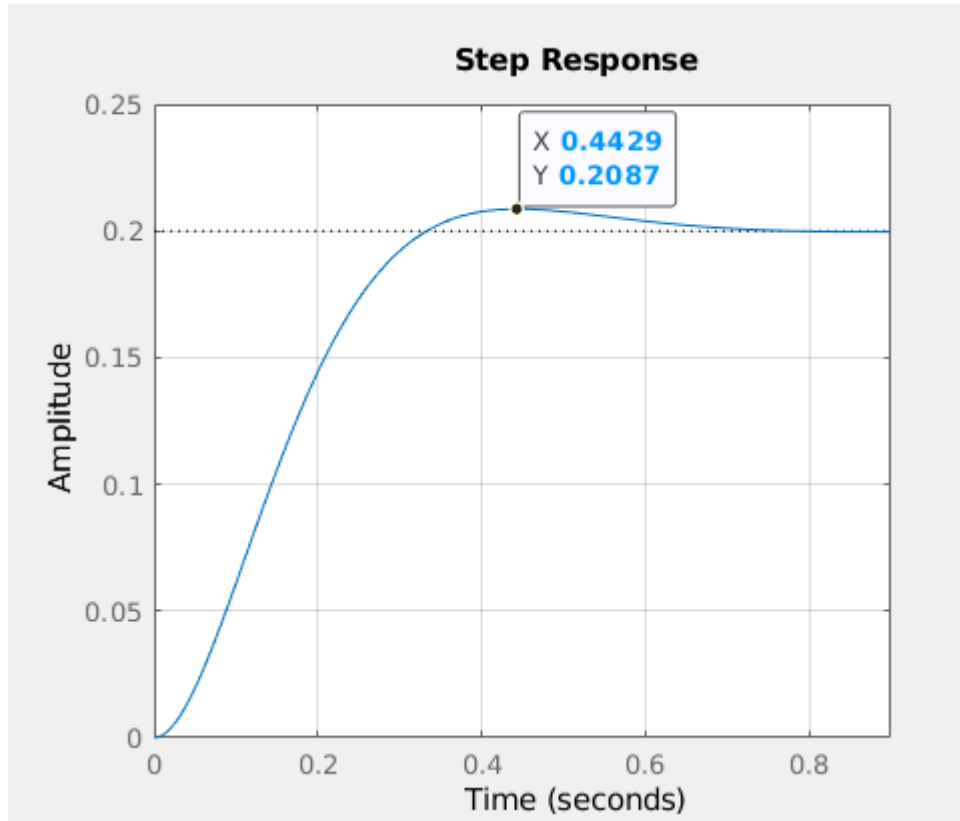
$$G_2(s) = K_{dc} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = K_{dc} \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_d^2}$$

$$T_c = \tau = 1/a$$

$K=11$ * assumed based on steady state settling time of T_s !

Second Order Model Estimation

Given an unit step response graph



2nd Order TF Model: **Three** Unknowns

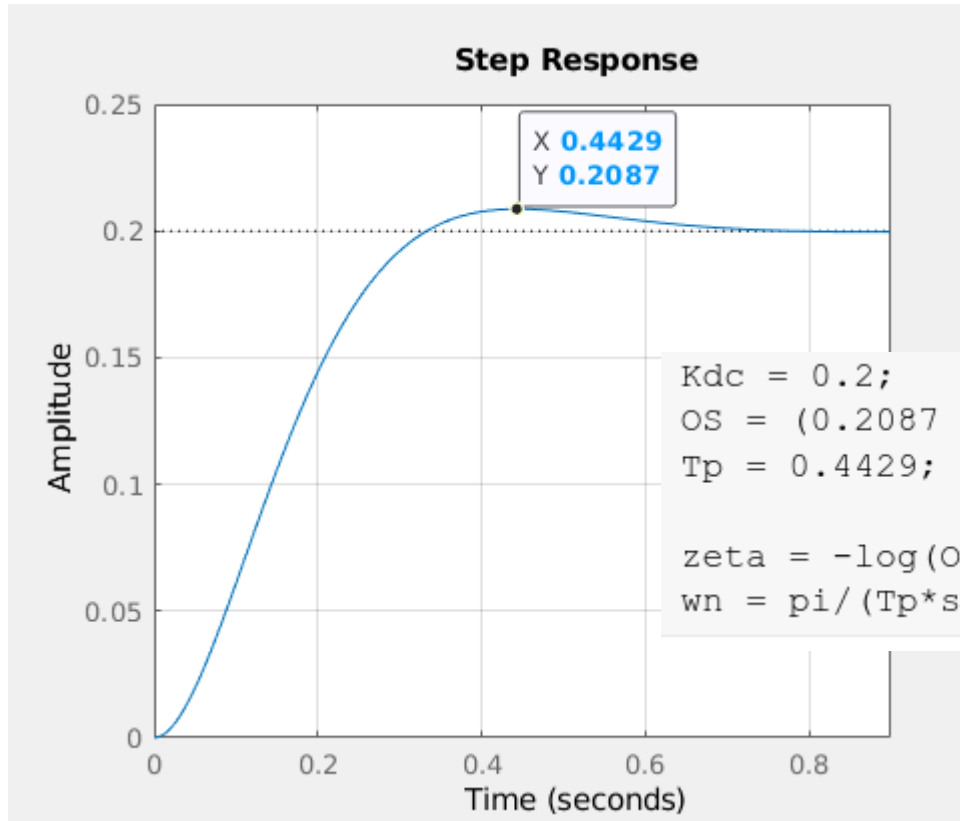
$$G_2(s) = K_{dc} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = K_{dc} \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_d^2}$$

Observable Metrics: **Four** Equations
Only use three

```
Kdc = 0.2;  
OS = (0.2087 - 0.2) / 0.2;  
Tp = 0.4429;
```

Second Order Model Estimation

Given an unit step response graph



2nd Order TF Model: **Three** Unknowns

$$G_2(s) = K_{dc} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = K_{dc} \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_d^2}$$

Observable Metrics: **Four** Equations
Only use three

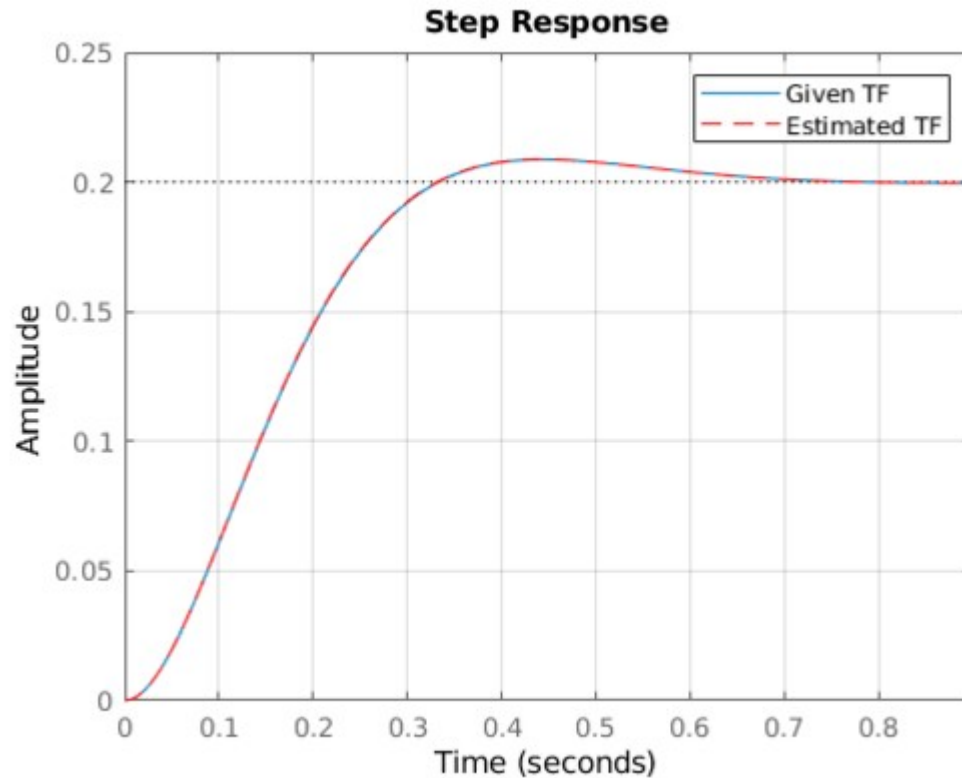
```
Kdc = 0.2;  
OS = (0.2087 - 0.2) / 0.2;  
Tp = 0.4429;  
  
zeta = -log(OS) / sqrt(pi^2 + log(OS)^2)  
wn = pi / (Tp * sqrt(1 - zeta^2))
```

zeta = 0.7064

wn = 10.0208

Second Order Model Estimation

Given an unit step response graph



2nd Order TF Model: **Three** Unknowns

$$G_2(s) = K_{dc} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = K_{dc} \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_d^2}$$

Observable Metrics: **Four** Equations
Only use three

```
G = tf(Kdc*wn^2,[1, 2*zeta*wn, wn^2])  
hold on  
step(G,'r--')  
legend('Given TF','Estimated TF')
```

Performance Metrics: Analytic vs Numeric

Analytic Solution:

Exact, but only for canonical second-order model

4.20

$$a) T(s) = \frac{16}{s^2 + 3s + 16}$$

$$\omega_n^2 = 16 \Rightarrow \omega_n = 4$$

$$3 = 2\zeta\omega_n = 2\zeta \cdot 4$$

$$\zeta = \frac{3}{8} = 0.375$$

$$\%OS = e^{-(2\pi/\sqrt{1-\zeta^2})} \times 100 = 28.06\%$$

$$\omega_n T_r = 1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1 = 1.424$$

$$T_r = 0.356s$$

$$\zeta = 0.375 \quad \omega_n = 4 \quad T_s = 2.667s \quad T_p = 0.8472s \quad T_r = 0.356s \quad \%OS = 28.06\%$$

$$T_s = \frac{4}{\zeta\omega_n} = \frac{4}{0.375 \cdot 4} = 2.667s$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} =$$

$$a) \zeta = 0.375$$

$$\omega_n = 4.00$$

$$T_s = 2.6570s$$

$$T_p = 0.8596s$$

$$T_r = 0.3569s$$

$$\%OS = 28.0255\%$$

Numerical Solution:

Estimate, but for any timeseries.

I am confused ~~where~~ where the T_p difference is coming from.
MATLAB vs calculated in 4.20 differ by $\sim 1\%$

I am getting similar error on later (b/c) with the stepinfo fun.

Compare Analytical and Numerical

4.20: Analytical Solution

For each of the following **second-order systems**, find performance metrics, e.g., peak-time T_p .

$$\text{a) } T(s) = \frac{16}{s^2 + 3s + 16}$$

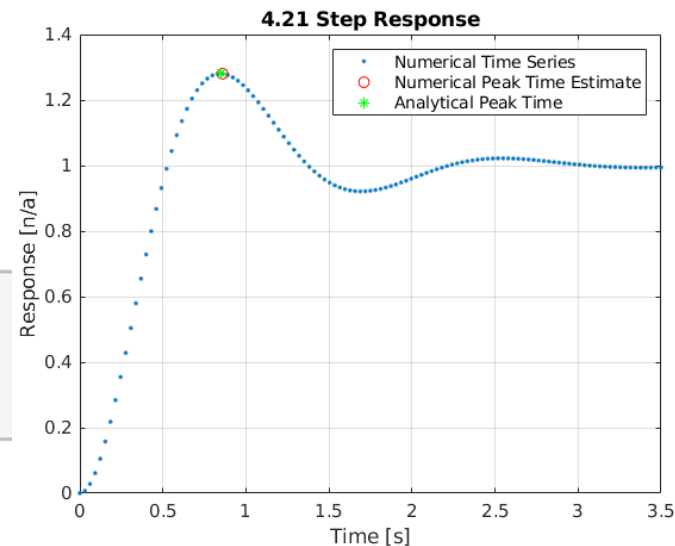
$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

```
omega_n = sqrt(16);  
zeta = 3 / (2*(omega_n));  
Tp_analytic = pi / (omega_n * sqrt(1-zeta^2))
```

4.21: Numerical Solution

This method is not limited to just our canonical second-order system model!

```
T = tf(16, [1 3 16]);  
Tp_numeric = stepinfo(T).PeakTime  
peak_numeric = stepinfo(T).Peak;
```



Compare Analytical and Numerical

4.20: Analytical Solution

For each of the following **second-order systems**, find perfc

$$\text{a) } T(s) = \frac{16}{s^2 + 3s + 16}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

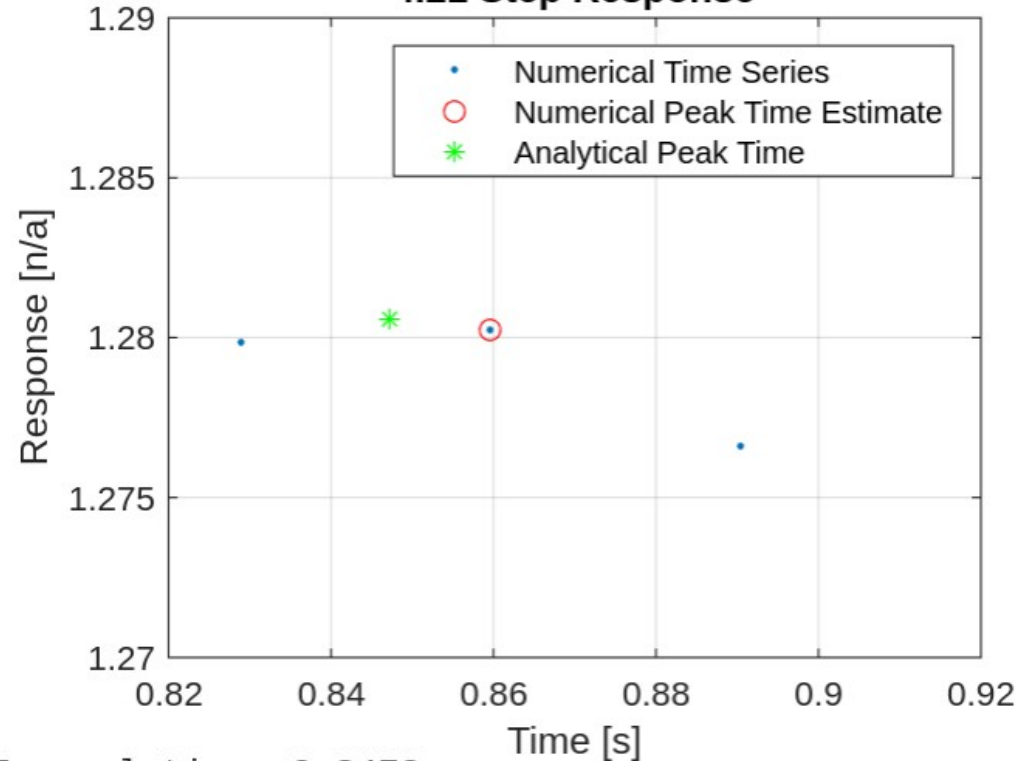
```
omega_n = sqrt(16);  
zeta = 3 / (2*(omega_n));  
Tp_analytic = pi / (omega_n * sqrt(1-zeta^2));
```

4.21: Numerical Solution

This method is not limited to just our canonical second-order

```
T = tf(16, [1 3 16]);  
Tp_numeric = stepinfo(T).PeakTime  
peak_numeric = stepinfo(T).Peak;
```

4.21 Step Response



$Tp_{analytic} = 0.8472$

$Tp_{numeric} = 0.8596$

22-1

4.20 a.) $T(s) = \frac{16}{s^2 + 3s + 16}$

GENERAL FORM: $G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

COULDN'T FIND THIS FORMULATION IN TEXT BOOK

$\omega_n^2 = 16 \Rightarrow \boxed{\omega_n = 4 \text{ rad/s}}$

$2\zeta\omega_n = 3 \Rightarrow \boxed{\zeta = 0.375}$

$T_s = \frac{4}{\zeta\omega_n} = \boxed{2.67 \text{ s}}$

$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = \boxed{0.85 \text{ s}}$

$\omega_n T_r = 1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1$
 $= \boxed{0.36 \text{ s}}$

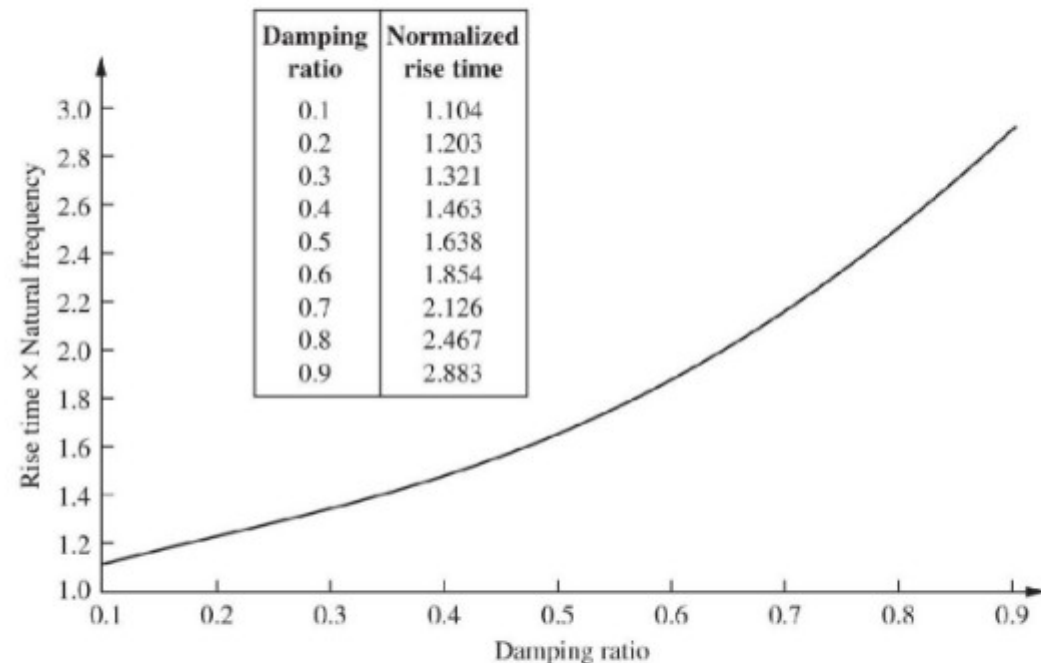
$\zeta = 0.375$

$\boxed{T_r = 0.3569 \text{ s}}$

used matlab hand calc is cumbersome for T_r

Rise Time of Underdamped Second Order System Step Response

According to the Nise textbook (7th Ed.) "a precise analytical relationship between rise time and damping ratio cannot be found." Instead a numerical solution is provided as represented by Nise FIGURE 4.16



and the following footnote:

Figure 4.16 can be approximated by the following polynomials

$$\omega_n T_r = 1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1 \quad (\text{max. error less than } 1/2\% \text{ for } 0 < \zeta < 0.9)$$

Sign Convention

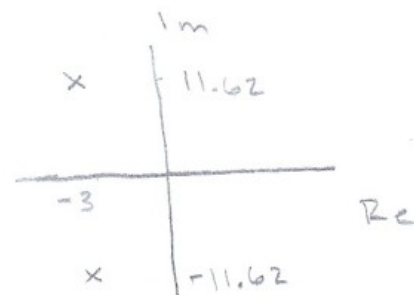
In the book, the phase shift is $\cos(\omega t - \phi)$ for underdamped, but it's $\cos(\omega t + \phi)$ in the solution.

$$c(t) = K[1 - A e^{-\sigma_d t} \cos(\omega_d t - \phi)]$$

$$c(t) = K[1 - A \cos(\omega_1 t - \phi)]$$

21-1

b) $T(s) = \frac{20}{s^2 + 6s + 144}$ $K = 0.139$
 poles = $-3 \pm 11.62j$
 $\leftarrow$ roots = $-3 \pm 11.62j$ $\uparrow$ $\xi \omega_n \pm j \omega_d$



$\omega_d = 11.62$
 $\zeta = 0.25$
 $\omega_n = 12$

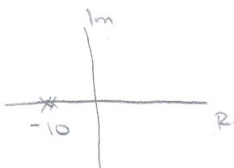
$$c(t) = 1.67 e^{-3t} \sin(11.62t)$$

2 complex ^{Re & Im} conjugates $\therefore$ Underdamped

Not the book answer?

d. Poles: $(-3 + j3\sqrt{15})$, $(-3 - j3\sqrt{15})$; $c(t) = A + Be^{-3t} \cos(3\sqrt{15} t + \phi)$; underdamped.

f) $T(s) = \frac{s+5}{(s+10)^2}$ zero = -5
 poles = 2 @ -10
 $K=1$ $\leftarrow$ real repeating roots $\therefore$ critically damped



$$c(t) = 1 - Ae^{-10t} - Bte^{-10t}$$

Why do the solutions have positive constants (A & B) while the general forms in the text have negative?

f. Poles: -10, -10; Zero: -5; $c(t) = A + Be^{-10t} + Cte^{-10t}$; critically damped.

$$\begin{aligned} c(t) &= K[1 - Ae^{-\sigma_1 t} - Be^{-\sigma_2 t}] \\ c(t) &= K[1 - Ae^{-\sigma_d t} \cos(\omega_d t - \phi)] \\ c(t) &= K[1 - A \cos(\omega_1 t - \phi)] \\ c(t) &= K[1 - Ae^{-\sigma_1 t} - Bte^{-\sigma_1 t}] \end{aligned}$$

①. 4.3 (Nice)

→ see MATLAB plots below. (a & b).

★ Is step response plot in time domain
always? ★

3) 4.24) a) $T(s) = \frac{16}{s^2 + 3s + 16}$

$[Z, \omega_n, T_s, T_p, T_r, \%OS]$

$$\omega_n = 4$$

$$Z = \frac{3}{2(4)} = 0.375$$

$$T_s = \frac{4}{(0.375)(4)} = 2.67 \text{ sec}$$

$$T_p = \frac{\pi}{(4)\sqrt{1-0.375^2}} = 0.817 \text{ sec}$$

$$K = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Does this always apply?

c) $T_s = 7 \text{ s} = \frac{4}{\zeta\omega_n} \rightarrow \zeta\omega_n = \frac{4}{7} = 0.571$

$T_p = 3 \text{ s} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} \rightarrow \omega_n\sqrt{1-\zeta^2} = \frac{\pi}{3} = 1.047$

poles ... ? formula is $-\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1}$, but we have $\omega_n\sqrt{1-\zeta^2} \dots ?$

3 Nise 4.20 and 4.21 Find $\zeta, \omega_n, T_s, T_p, T_r, \%OS$
4.20) a) $T(s) = \frac{16}{s^2 + 3s + 16}$ $\omega_n = 4$

$\Rightarrow 2\zeta\omega_n = 3$ *not sure where eqn comes from*

$$K_{DC} \frac{\Re^2 + \Im^2}{(s + \Re)^2 + (\Im)^2} = K_{DC} \frac{(\zeta \omega_n)^2 + (\omega_d)^2}{(s + \zeta \omega_n)^2 + (\omega_d)^2} = K_{DC} \frac{\omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

$$a) T(s) = \frac{16}{s^2 + 3s + 16}$$

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

$$T_r: \omega_n T_r = (1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1) = 1.4238$$

$$\therefore T_r = 0.356s$$

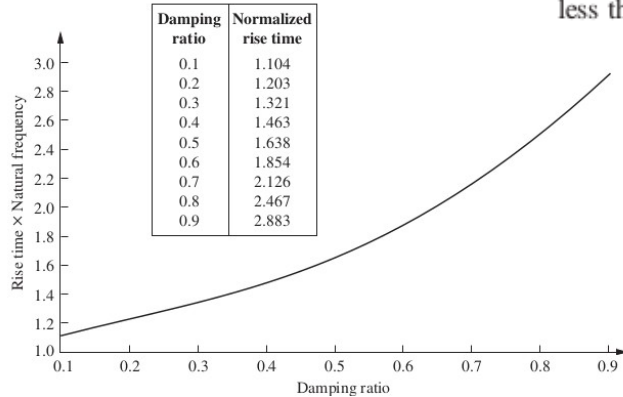
Can you please explain this equation for T_r

Evaluation of T_r

A precise analytical relationship between rise time and damping ratio, ζ , cannot be found. However, using a computer and Eq. (4.28), the rise time can be found. We first designate $\omega_n t$

*struggling w/ $T_r \dots (1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1) = 1.4238$

⁵ Figure 4.16 can be approximated by the following polynomials: $\omega_n T_r = 1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1$ (maximum error less than $\frac{1}{2}\%$ for $0 < \zeta < 0.9$), and $\zeta = 0.115(\omega_n T_r)^3 - 0.883(\omega_n T_r)^2 + 2.504(\omega_n T_r) - 1.738$ (maximum error less than 5% for $0.1 < \zeta < 0.9$). The polynomials were obtained using MATLAB's **polyfit** function.



$$\omega_n^2 = 16 \text{ r/s} \Rightarrow \omega_n = 4$$

$$2\zeta\omega_n = 3 \Rightarrow 2\zeta 4 = 3 \Rightarrow \zeta = 0.375$$

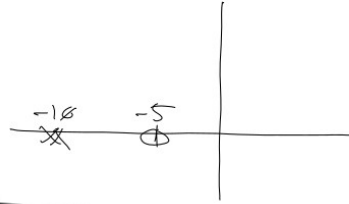
FIGURE 4.16 Normalized rise time versus damping ratio for a second-order underdamped response

$$f) T(s) = \frac{(s+5)}{(s+10)^2}$$

Zero's = -5
pole's = -10, -10

critical damp

$$T(t) = (1) \left[1 - e^{-10t} - e^{-10t} \right]$$



Do you have to include A/B in front of
Equation components?
ie: Ae^{-10t} vs. e^{-10t}

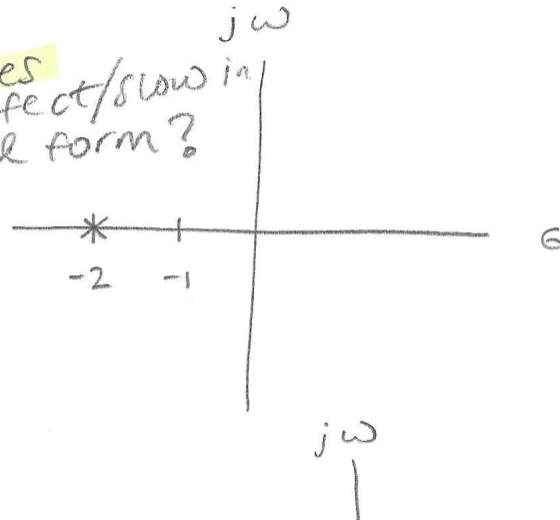
②. 1.8 (Nice)

a.

$$T(s) =$$

$$\frac{2}{s+2}$$

How does
this affect/slow in
the general form?



$$p = -2$$

$$\rightarrow c(t) = K \left[1 - e^{-2t} \right]$$

(b). $C_{max} = 13.75$
 $C_{final} = 10.625$
 from graph

$$\%OS = \frac{13.75 - 10.625}{10.625} \times 100$$

$$= 29.41$$

$$\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}$$

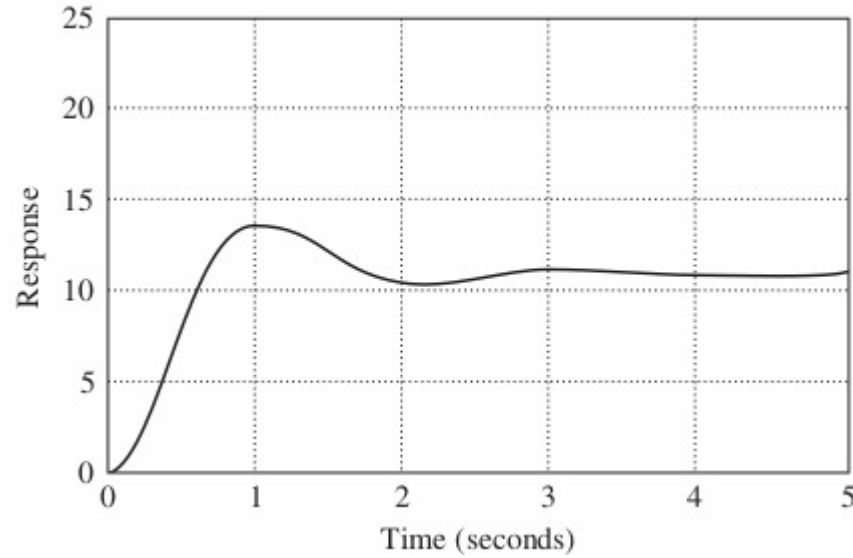
$$\zeta = 0.363$$

$$T_s = 2.375$$

$$= \frac{4}{\zeta \omega_n} \rightarrow \omega_n = 4.64$$

$$G(s) = \frac{21.53}{s^2 + 3.375s + 21.53}$$

Answer key has 160.95?
 uses $\frac{K}{\omega_n^2} = C_{final}$... where
 does this come from?



(4.42) to find $\omega_n = 3.82$. Thus, $G(s) = \frac{K}{s^2 + 2\zeta\omega_n s + \omega_n^2}$. Since $C_{final} = 11.03$, $\frac{K}{\omega_n^2} = 11.03$.

Hence, $K = 160.95$. Substituting all values,

$$G(s) = \frac{160.95}{s^2 + 3.056s + 14.59}$$

$$K_{DC} \frac{\Re^2 + \Im^2}{(s + \Re)^2 + (\Im)^2} = K_{DC} \frac{(\zeta \omega_n)^2 + (\omega_d)^2}{(s + \zeta \omega_n)^2 + (\omega_d)^2} = K_{DC} \frac{\omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$